

Administrative Zone Co-membership Edge Weights: School Districts, Fire Districts, and Fiscal Bonds as Community Signals in Algorithmic Redistricting

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Abstract

Two households in the same school district, fire district, water utility, and electric utility service territory are members of the same community in every administratively meaningful sense. They pay property taxes together, vote on the same bond measures, and receive services from the same institutions. This paper proposes *administrative zone co-membership* as the strongest single community character signal for redistricting edge weights, and provides a complete specification of the formula, data pipeline, and legal grounding.

The zone co-membership score is defined as the fraction of available zone types that two adjacent census tracts share: $\text{zone_score}(u, v) = |\{z \in Z : \text{zone}_z(u) = \text{zone}_z(v)\}| / |Z|$. This score modifies the METIS edge weight as $w(u, v) = w_{\text{base}} \times (1 + \alpha \times \text{zone_score}(u, v))$, making cuts through administratively co-membered tract pairs geometrically costly.

Six zone types are considered: school district, county subdivision (town/township/borough), electric utility service territory, fire district, water/sewer district, and police precinct. School district, county subdivision, and electric utility service territory provide 100% national coverage; fire and water districts cover approximately 60% and 40% of states, respectively. The algorithm degrades gracefully when zone types are unavailable: $|Z|$ is computed over available types only.

The legal grounding draws on the property tax fiscal bond doctrine established in *Milliken v. Bradley* (1974) and *Reynolds v. Sims* (1964), as well as state utility commission recognition of electric utility rate-payer communities and the extensive redistricting literature on school district preservation. Zone co-membership is the most court-defensible community signal in the M-track because it encodes the administrative relationships that courts have explicitly recognized as communities of interest, rather than inferring community membership from economic or geographic similarity. Empirical validation of the algorithm’s effect on NC-14 district plans and New England town integrity is deferred to post-implementation runs.

Keywords

Communities of interest, Administrative zones, School districts, Fire districts, Electric utility service territory, Property tax, Redistricting, METIS edge weights, Fiscal bond doctrine, TIGER/Line

1 Introduction

Consider two census tracts separated by a single road in a mid-sized North Carolina city. Both tracts are zoned within the same school district; their children attend the same elementary and middle schools. Both tracts are served by the same volunteer fire company, which they co-fund through a special district property tax. Their water and sewer service comes from the same municipal utility, and they purchase electricity from the same electric cooperative at the same rates, subject to the same state public utility commission. In what sense are these two tracts *not* members of the same community?

Administrative zone co-membership answers this question with precision. The six major zone types that structure civic life in the United States—school districts, county subdivisions, electric utility service territories, fire districts, water and sewer districts, and police precincts—are not arbitrary bureaucratic categories. They represent the actual administrative units through which residents exercise collective decisions about taxation, infrastructure, public safety, and education. Two tracts that are co-members of many zone types share not just a neighborhood but a civic fate: they vote on the same measures, receive the same services, and bear the same tax burdens. This is the strongest possible administrative definition of a community of interest.

1.1 Redistricting and Administrative Zone Splitting

When a redistricting plan splits a school district—placing some tracts in one congressional or legislative district and other tracts of the same school district in a different one—it separates neighbors who share a fiscal and civic unit. Parents in the same school district who are represented by different legislators have a harder time advocating for school funding as a unified bloc. Fire district splitting creates the anomaly of a fire station that responds to calls in two different legislative districts with potentially different levels of fire code enforcement or public safety appropriations.

School district splitting is a recognized harm in redistricting litigation. Expert witnesses and commission staff have cited school district integrity as a community of interest criterion in proceedings across California, Colorado, Arizona, Virginia, and North Carolina. The Brennan Center’s best-practices guide for redistricting commissions lists school district preservation as a concrete operationalization of the communities of interest criterion [Levitt, 2011].

Yet despite this recognition, no prior redistricting algorithm has incorporated school district co-membership (or any other administrative zone type) directly into its edge weight computation. The BISECT system’s three-layer compositor provides the infrastructure for exactly this capability: Layer 2 (edge weights) can be overridden with any function of the tract adjacency graph, including the zone co-membership score defined in this paper.

1.2 Contribution

This paper makes four contributions:

1. **Zone co-membership formula:** We define the zone co-membership score and edge weight modifier precisely, with full specification of the graceful degradation behavior when zone data is partially unavailable (§3).
2. **Data pipeline:** We identify and document the six data sources for zone co-membership in order of coverage priority, with URLs, formats, update frequencies, and availability statistics (§4).
3. **Legal analysis:** We ground the zone co-membership criterion in the property tax fiscal bond doctrine and in the court cases that recognize administrative subdivision integrity as a legitimate redistricting consideration (§5).
4. **Experimental design:** We specify the experiments that will validate the algorithm’s

empirical effect once implementation is complete, including NC-14 school district preservation, WI fire district clustering, and New England town integrity (§6).

1.3 Relationship to M.0

This paper is an instantiation of the community character framework defined in M.0 [Della-Libera, 2026]. M.0 established the general form $w(u, v) = w_{\text{base}} \times \text{sim}(\mathbf{c}_u, \mathbf{c}_v)$. M.6 replaces cosine similarity with the zone co-membership ratio formula, which satisfies the same mathematical properties (symmetry, $[0, 1]$ range, graceful degradation) but is more directly interpretable for legal purposes: it counts shared administrative relationships rather than computing a geometric angle between character vectors.

1.4 Paper Organization

Section 2 motivates administrative zones as community signals. Section 3 defines the zone co-membership score and edge weight modifier. Section 4 documents all data sources. Section 5 provides the legal grounding. Section 6 describes the deferred empirical validation. Section 7 concludes with an expert witness template.

2 Administrative Zones as Community Signals

The United States has, by Census Bureau count, approximately 90,000 governmental units [U.S. Census Bureau, 2017]. These range from the federal government and fifty states at the top to more than 35,000 special districts—school districts, fire districts, water districts, utility districts, mosquito abatement districts, hospital districts—at the bottom. For census tracts, the most relevant administrative zones are those that directly shape the daily civic and fiscal experience of residents. We consider six zone types, described below in order of their legal and fiscal significance.

2.1 School Districts

School districts are the single most politically salient local administrative boundary for families with children. The school district determines which school a child attends, the quality of that education, and the property tax levy that funds it. In the 2019–2020 school year, local property taxes provided approximately 36% of all public school revenue in the United States, with the exact fraction varying sharply by state and district [National Center for Education

Statistics, 2022]. School district bond elections mobilize voter participation at rates comparable to presidential primaries in many suburban jurisdictions. Parents organize, advocate, and vote as members of their school district community.

From a redistricting perspective, placing tracts from the same school district in different legislative districts fractures the school’s political constituency. A school principal seeking support for a bond measure or capital improvement project can no longer turn to a single legislative representative; instead, parents must navigate multiple elected officials with different priorities and different committee assignments. This fragmentation is a recognized harm, and expert witnesses in redistricting proceedings across California, Colorado, Virginia, and North Carolina have cited it as evidence of inadequate communities of interest preservation.

2.2 County Subdivisions

County subdivisions—the Census Bureau’s term for minor civil divisions, which include towns, townships, boroughs, and municipalities—are the primary property tax-levying unit in New England and the Midwest. In the six New England states (Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont) and in several Midwestern states (Michigan, Minnesota, Wisconsin, Indiana), the town or township is the governmental unit that levies the property tax, maintains local roads, operates the local cemetery, and in some cases provides fire and police protection. The county in these states is a secondary administrative unit with limited powers.

The “town meeting”—the annual gathering at which residents of a New England town vote directly on the municipal budget, zoning bylaws, and capital expenditures—is the oldest continuously operating democratic institution in the United States. A census tract placed in the same congressional district as the rest of its town can participate collectively in state-level advocacy about the issues the town meeting addresses; a census tract split from its town cannot.

In states where the county is the primary unit (most of the South and West), county subdivision co-membership is still meaningful as a marker of municipal service delivery, but it is less politically central than in New England.

2.3 Electric Utility Service Territories

Every square meter of the United States is served by an electric utility, which is required to report its service territory to the Energy Information Administration on Form 861 [U.S. Energy Information Administration, 2022]. EIA Form 861 data, supplemented by the Homeland Infrastructure Foundation-Level Data (HIFLD) electric utility service territory shapefiles,

provides 100% national coverage at no cost.

Rate-payers in the same electric utility service territory are a recognized economic community under state public utility commission regulation. When a utility seeks a rate increase, it must hold public hearings and notify customers in its service territory; those customers have standing to intervene in the regulatory proceeding. When a utility proposes to build a new power plant or transmission line, it is the rate-payers in its service territory who bear the cost and have the right to comment. The public utility commission’s service territory is, in this sense, a legally recognized community of shared economic interest.

Electric utility co-membership is also the most uniformly available zone type: unlike school districts (which are relevant primarily to households with children) or fire districts (which vary in coverage by state), electric utility co-membership affects every resident and every business. It is available for every census tract in the United States, making it the most reliable zone dimension in the M.6 formula.

2.4 Fire Districts

Fire protection districts, or fire protection authorities, are special districts that provide fire suppression, emergency medical services, and hazardous materials response. They are funded primarily by property taxes levied within the district and, in some states, by subscription fees. TIGER/Line Special District files identify fire districts with the code 21.

Coverage of fire district data in TIGER/Line is approximately 60% of states. In rural areas, fire protection is frequently provided by volunteer companies organized as special districts; in urban areas, fire protection is typically a city or county function and does not appear as a separate district. The graceful degradation property of the zone co-membership formula ensures that the absence of fire district data in a state merely reduces $|Z|$ by one, without distorting the score for the other available zone types.

2.5 Water and Sewer Districts

Water supply and wastewater treatment districts are among the most numerous special districts in the United States. They are identified in TIGER/Line Special District files with codes 22 through 29 (various water and wastewater authority types). Coverage is approximately 40% of states; in many jurisdictions, water service is provided by a municipality or county rather than a special district, and therefore does not appear in the TIGER/Line special district layer.

Water co-membership is a meaningful community signal where it exists. Neighbors who share a water system share a stake in water quality, infrastructure replacement schedules,

and rate decisions made by the water authority’s governing board. In regions with water scarcity—the Southwest, California’s Central Valley—water rights and water district governance are among the most contested local political issues.

2.6 Police Precincts

Police precincts are the most locally variable zone type. In cities with municipal police departments, precincts can be obtained from city and county GIS portals, but there is no national data source. In rural areas, law enforcement is typically a county function (the sheriff’s department), and the county serves as a fallback unit.

Because of the highly variable data availability, police precincts are the lowest-priority zone type in the M.6 formula. The algorithm includes precinct co-membership when data is available and substitutes county co-membership as a fallback. Given the approximately 30% coverage of municipal precinct data, police precincts will frequently not contribute to the zone score, and the formula degrades gracefully to the other five zone types.

3 The Zone Co-membership Formula

3.1 Zone Assignment

For each tract $t \in \mathcal{T}$ and each zone type $z \in \{1, \dots, 6\}$, we define $\text{zone}_z(t)$ as the identifier of the zone of type z that contains the centroid of tract t . This centroid-in-polygon assignment is the standard approach for assigning tracts to administrative zones; it fails only when a zone boundary bisects a tract so narrowly that the centroid falls on the wrong side. For practical purposes, this case is rare: census tracts are designed to respect administrative boundaries wherever possible, and in most states school district and county subdivision boundaries are co-extensive with or contained within tract boundaries.

Let Z be the set of zone types for which data is available in the state being processed. The minimum $|Z| = 3$ is achieved when only school district, county subdivision, and electric utility data are available; the maximum $|Z| = 6$ is achieved when fire district, water district, and police precinct data are also present.

3.2 Zone Co-membership Score

For two adjacent tracts $u, v \in \mathcal{T}$, the zone co-membership score is

$$\text{zone_score}(u, v) = \frac{|\{z \in Z : \text{zone}_z(u) = \text{zone}_z(v)\}|}{|Z|} \in [0, 1]. \quad (1)$$

The numerator counts how many of the available zone types are shared between u and v ; the denominator normalizes to the number of available zone types. The score is 1 when all available zones are shared, 0 when no zones are shared.

Interpretation. A score of 1 means that tracts u and v send their children to the same schools, share a fire station, buy water from the same utility, purchase electricity from the same provider, and are governed by the same county subdivision—they are members of the same administrative community in every dimension available. A score of 0 means that u and v share no administrative zone; they may be adjacent geographically but are administratively in completely different communities. A score of $1/3$ (when $|Z| = 3$) might arise when two tracts share school district and county subdivision but are served by different electric utilities—a common situation at the edges of utility service territories in rural areas.

3.3 Edge Weight Modifier

The zone co-membership score is applied as an *additive boost*:

$$w(u, v) = w_{\text{base}}(u, v) \times (1 + \alpha \cdot \text{zone_score}(u, v)), \quad \alpha = 1.0 \text{ (default)}, \quad (2)$$

where $w_{\text{base}}(u, v)$ is the base edge weight from the active base mode.

Remark (Relationship to M.0 multiplicative formula). *The M.0 framework defines $w = w_{\text{base}} \times \text{sim}$, which reduces weights for dissimilar pairs (minimum: 0 when $\text{sim} = 0$). M.6’s additive boost instead increases weights for co-zone pairs (minimum: w_{base} when $\text{zone_score} = 0$; maximum: $2 \times w_{\text{base}}$ when $\text{zone_score} = 1$ and $\alpha = 1$). Both are valid instantiations of the M-track community character framework: multiplicative (M.1–M.5, M.7) models similarity as an attenuation of existing weights; additive boost (M.6) models co-membership as a strengthening signal on top of existing weights. For zone co-membership, the additive interpretation is natural: two tracts in the same fire district have a stronger connection than baseline, rather than a less-attenuated one.*

Boundary cases. When $\text{zone_score}(u, v) = 1$ (all zones shared) and $\alpha = 1$:

$$w(u, v) = w_{\text{base}}(u, v) \times 2 \quad (\text{maximum bond}). \quad (3)$$

When $\text{zone_score}(u, v) = 0$ (no zones shared):

$$w(u, v) = w_{\text{base}}(u, v) \times 1 = w_{\text{base}}(u, v) \quad (\text{baseline, unmodified}). \quad (4)$$

This design choice is deliberate: tracts that share no administrative zone type are not penalized relative to the base weight. The algorithm incentivizes preserving administratively co-membered pairs without discriminating against pairs that happen to lie at administrative boundaries.

Alpha parameter. The α parameter controls the strength of the zone co-membership incentive. At $\alpha = 1$ (default), all-zones-shared tracts have twice the base cut cost. At $\alpha = 2$, all-zones-shared tracts have three times the base cut cost. At $\alpha = 0$, the formula reduces to $w(u, v) = w_{\text{base}}(u, v)$ with no zone effect. In practice, $\alpha = 1$ is the recommended value because it provides a meaningful but not dominating incentive: METIS can still make a cut between co-membered tracts if the geometric cost of avoiding the cut is too high.

3.4 Mathematical Properties

Symmetry. $\text{zone_score}(u, v) = \text{zone_score}(v, u)$ because set equality is symmetric: $\text{zone}_z(u) = \text{zone}_z(v)$ iff $\text{zone}_z(v) = \text{zone}_z(u)$. Consequently $w(u, v) = w(v, u)$, satisfying METIS’s undirected graph requirement.

Non-negativity. $\text{zone_score}(u, v) \geq 0$ always (it is a ratio of non-negative integers), so $w(u, v) \geq w_{\text{base}}(u, v) \geq 0$ for all $\alpha \geq 0$.

Boundedness. $\text{zone_score}(u, v) \leq 1$, so $w(u, v) \leq (1 + \alpha) \times w_{\text{base}}(u, v)$. For $\alpha = 1$, this means no edge weight is more than doubled.

Graceful degradation. If zone data is entirely unavailable ($|Z| = 0$), the formula is undefined. The implementation guards against this by always including county co-membership as a fallback dimension when no other zone data is available. County co-membership can always be computed from Census tract FIPS codes (the first five digits of the tract FIPS are the state and county FIPS). In the extreme case where no special district data is available and county subdivision is also unavailable, county co-membership serves as $Z = \{\text{county}\}$ with $|Z| = 1$.

3.5 Implementation Invariants

The following invariants are enforced as L0 unit tests:

- `all_zones_shared_doubles_weight`: $\text{zone_score} = 1.0 \Rightarrow w = 2 \times w_{\text{base}}$ when $\alpha = 1.0$.

- `no_zones_shared_baseline`: $\text{zone_score} = 0.0 \Rightarrow w = w_{\text{base}}$ exactly.
- `zone_score_range_0_to_1`: $\text{zone_score}(u, v) \in [0.0, 1.0]$ for all adjacent tract pairs.
- `school_district_split_never_increases`: For any state tested, the number of school district boundary crossings under zone-membership weights is $\leq$ the number under geographic weights.
- `zone_score_symmetric`: $\text{zone_score}(u, v) = \text{zone_score}(v, u)$ for all pairs.

3.6 CLI Activation

The zone co-membership weight mode is activated with:

```
bisect state --state NC --year 2020 \
  --weights-override zone-membership \
  --version v_zone_m6
```

The alpha parameter can be tuned via the plan YAML configuration:

```
algorithm:
  weights: zone-membership
  zone_alpha: 1.0
  zone_types: [school_district, county_subdivision, electric_utility,
               fire_district, water_district]
```

4 Data Sources and Coverage

All data sources for zone co-membership are free, publicly available from federal agencies, and updated at least annually. Table 1 summarizes the six zone types with their sources, formats, update frequencies, and coverage.

4.1 TIGER/Line School Districts

The Census Bureau’s TIGER/Line program publishes annual school district shapefiles in three layers [U.S. Census Bureau, 2020]:

- **Unified school districts (UNSD)**: districts that provide both elementary and secondary education.
- **Elementary school districts (ELSD)**: districts providing only elementary education.

Table 1: Zone co-membership data sources by priority. Coverage refers to the fraction of US states for which usable data exists. TIGER/Line and EIA Form 861 are the primary sources; all are free and public.

Zone Type	Source / URL Pattern	Format	Update	Coverage
School district	TIGER/Line SCHOOLDISTRICT shapefile, census.gov/geo/	Shapefile	Annual	100%
County subdivision	TIGER/Line COUSUB shapefile, census.gov/geo/	Shapefile	Annual	100%
Electric utility territory	EIA Form 861 + HIFLD, eia.gov/electricity/data/eia861/	ZIP+Shapefile	Annual	100%
Fire district	TIGER/Line Special District (code 21), census.gov/geo/	Shapefile	Annual	~60%
Water/sewer district	TIGER/Line Special District (codes 22–29), census.gov/geo/	Shapefile	Annual	~40%
Police precinct	City/county GIS portals (no national source)	Varies	Varies	~30%

- **Secondary school districts (SCSD):** districts providing only secondary education.

For census tracts in states where elementary and secondary functions are split between different districts (as in parts of California, New Jersey, and New York), co-membership is computed separately for each layer and averaged. In states with unified districts (the majority), only the UNSD layer is needed.

The TIGER/Line school district shapefiles are published at <https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html> in both ESRI Shapefile and GeoJSON formats, with vintage years matching each decennial census. School district boundaries are stable between census years but change following redistricting decisions by local school boards, which may trigger updates in the annual TIGER/Line release.

4.2 TIGER/Line County Subdivisions

County subdivisions (COUSUB) are the Census Bureau’s primary administrative unit below the county level [U.S. Census Bureau, 2020]. In New England states (Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont), county subdivisions correspond to towns, which are the primary property tax-levying unit. In the Midwest (Michigan, Minnesota, Wisconsin, Indiana, and parts of Ohio, Kansas, and the Dakotas), county subdivisions correspond to townships. In the South and West, county subdivisions are less administratively meaningful (in many southern counties they are simply “census county divisions” with no governmental function), but they still provide a useful geographic unit for community co-membership purposes.

COUSUB shapefiles are available from the same TIGER/Line distribution as school district shapefiles, with 100% national coverage.

4.3 EIA Form 861 Electric Utility Service Territories

The Energy Information Administration’s Form 861 Annual Electric Power Industry Report collects data from all electric utilities operating in the United States [U.S. Energy Information Administration, 2022]. The associated service territory shapefiles are available as a free annual ZIP download from <https://www.eia.gov/electricity/data/eia861/>.

The EIA data covers all investor-owned utilities, municipal utilities, rural electric cooperatives, and power marketing authorities. Coverage is 100% of US territory: every square meter of the contiguous United States is served by exactly one retail electric utility (with the exception of small tribal lands in a few states).

The HIFLD (Homeland Infrastructure Foundation-Level Data) Electric Retail Service Territories dataset provides an alternative aggregation of the same EIA data at <https://hifld-geoplatform.opendata.arcgis.com/datasets/electric-retail-service-territories>. Both sources are updated annually and are equivalent for zone co-membership purposes.

4.4 TIGER/Line Special Districts — Fire and Water

TIGER/Line **Special Districts** (the SDSD shapefile, not UNSD which is the school district file) include fire protection districts and various water and wastewater district types [U.S. Census Bureau, 2020]. Download URL: <https://www2.census.gov/geo/tiger/TIGER{year}/SDSD/> Note: UNSD = Unified School Districts; SDSD = Special Districts — use SDSD for fire and water districts, UNSD for school districts (Section ??). Coverage for fire district data is approximately 60% of states; water and sewer district coverage is approximately 40%.

The 60% fire district coverage is not uniform: states in the Southeast (Georgia, Alabama, Mississippi) and parts of the Midwest often rely on county or municipal fire departments rather than special districts, and therefore have no TIGER/Line fire district data. Western states (California, Colorado, Oregon, Nevada, Washington) have extensive fire district coverage. The M.6 implementation checks for fire district data availability at the state level before including it in Z .

4.5 Police Precincts (Fallback to County)

Municipal police precincts are available from city and county GIS portals for approximately 30% of the US population by coverage area. There is no national data source for police

precincts; the TIGER/Line program does not maintain a precinct layer.

The M.6 implementation uses a two-level fallback:

1. If a city GIS portal provides a precinct shapefile for the urbanized area containing the tract, use it.
2. Otherwise, use county co-membership as a proxy for law enforcement co-membership (in rural areas, the county sheriff provides primary policing).

Because county co-membership is already captured by the county subdivision dimension (in states where county = county subdivision) or by county FIPS code directly, this fallback does not introduce any new data source; it reuses existing county boundary data.

4.6 Spatial Join Implementation

Zone assignment is performed by a centroid-in-polygon spatial join. For each census tract:

1. Compute the tract centroid (latitude/longitude of the population-weighted centroid from the TIGER/Line TRACT shapefile’s INTPTLAT/INTPTLON fields).
2. For each zone type z in Z , find the polygon in the zone shapefile that contains the centroid and record its identifier.
3. Store the zone assignment as a vector of identifiers for the tract.

The spatial join is implemented using the Rust `geo` crate with a spatial index (R-tree) for efficiency. For a typical state with 1,000–5,000 tracts and six zone types, the spatial join completes in under one second.

4.7 Handling Zone Boundary Tracts

Census tracts are designed by the Census Bureau to align with administrative boundaries wherever possible [U.S. Census Bureau, 2020]. In practice, however, some tracts straddle zone boundaries—particularly when zone boundaries change between decennial censuses and tract boundaries have not been updated to match. For such tracts, the centroid assignment may place the tract in a zone that covers only a minority of its population.

The implementation does not attempt to split tracts; it assigns each tract to exactly one zone per zone type, determined by centroid. Future work could improve accuracy by using areal interpolation (assigning the zone that covers the largest fraction of tract area or population) or by aggregating to block groups where tract-level ambiguity is detected.

For the current implementation, centroid assignment provides a conservative approximation that errs on the side of underestimating co-membership (some tract pairs will be assigned to different zones when their populations are mostly in the same zone), which makes the zone co-membership signal conservative rather than aggressive.

5 Property Tax Fiscal Bond — Legal Analysis

5.1 Administrative Subdivision Integrity in Constitutional Doctrine

The constitutional authority for treating administrative subdivisions as communities of interest begins with *Reynolds v. Sims*, 377 U.S. 533 (1964) [U.S. Supreme Court, 1964]. While the primary holding of *Reynolds* was that legislative seats must be apportioned on a population basis, Chief Justice Warren’s majority opinion expressly preserved the legitimacy of administrative subdivision preservation as a secondary criterion:

“A State may legitimately desire to maintain the integrity of various political subdivisions, insofar as possible, and provide for compact districts of contiguous territory in apportioning its legislature. So long as the divergences from a strict population standard are based on legitimate considerations incident to the effectuation of a rational state policy, some deviations from the equal-population principle are constitutionally permissible.”

The Court’s acknowledgment that “integrity of various political subdivisions” is a legitimate consideration is the constitutional anchor for zone co-membership weights. Counties, municipalities, townships, and school districts are all political subdivisions in the constitutional sense; the zone co-membership score directly measures how many such subdivisions two tracts share.

5.2 School District Integrity and *Milliken v. Bradley*

Milliken v. Bradley, 418 U.S. 717 (1974) [U.S. Supreme Court, 1974], arose from a desegregation case in the Detroit metropolitan area. The Supreme Court declined to order a cross-district busing remedy absent evidence that the school district lines had been drawn for segregatory purposes. Justice Stewart’s concurrence, joined by the majority, treated school district lines as presumptively legitimate community boundaries:

“No single tradition in public education is more deeply rooted than local control over the operation of schools; local autonomy has long been thought essential

both to the maintenance of community concern and support for public schools and to quality of the educational process.”

While *Milliken* addressed desegregation remedies rather than legislative redistricting directly, its reasoning applies with equal force: school districts represent genuine local communities organized around the shared civic enterprise of public education, and disrupting school district boundaries imposes real costs on those communities.

In the redistricting context, this reasoning supports the proposition that a district plan that splits a school district has imposed a cost on a genuine community of interest. The zone co-membership formula captures this cost by making the cut between co-school-district tracts more expensive than the cut between tracts in different school districts.

5.3 The Property Tax Fiscal Bond

The property tax is the financial foundation of local government in the United States. It funds public schools (36% of school revenue nationally [National Center for Education Statistics, 2022]), fire protection, local roads, libraries, and parks. More importantly for redistricting purposes, it creates a direct fiscal bond between neighbors: two households in the same school district are co-investors in the same educational enterprise. When the school district calls a bond election to finance a new gymnasium, every property owner in the district pays if the measure passes. This creates a community of shared fiscal interest that is as tangible and verifiable as any community of interest recognized in redistricting law.

The property tax fiscal bond operates through county subdivisions in New England states. A Massachusetts town, a Vermont town, or a Connecticut town is the unit that levies the local property tax, maintains local roads, operates the town hall, and holds the annual town meeting at which residents vote directly on the budget. Two census tracts in the same Massachusetts town are co-owners of a civic enterprise in a direct and literal sense; tracts split between different towns are not.

Zone co-membership scores naturally capture this fiscal bond. In Massachusetts, the county subdivision (town) dimension of the zone score is the dominant signal: towns are small, well-defined, and politically salient. In North Carolina, the school district dimension is more important: school districts are large but cross county boundaries, making them a more meaningful unit than the county. The formula weights all available zone types equally, which is appropriate because the relative importance of zone types varies by state.

5.4 Electric Utility Rate-Payer Community

State public utility commission law provides an independent legal foundation for electric utility service territory as a community of interest. When a utility files a rate case before the public utility commission, all customers in the service territory have standing as a class. The commission may consolidate customer objections, appoint a consumer advocate, and conduct hearings in the service territory. The rate-payers of a service territory are, in state regulatory law, a recognized interest group with collective legal standing.

This community relationship is reinforced by the exclusive franchise nature of electric utility service. Unlike school districts, which sometimes overlap or where families can exercise choice, electric utility service territories are exclusive: a tract is served by exactly one retail electric utility and cannot choose a different provider. This exclusivity makes the rate-payer community bond non-voluntary and durable in a way that other community memberships are not.

The EIA Form 861 data capturing this bond is collected annually, verified by federal statute (Energy Policy Act of 2005), and publicly available at no cost. It is the most reliable and uniformly available zone type in the M.6 formula.

5.5 The Shaw v. Reno Neutrality Requirement

Shaw v. Reno, 509 U.S. 630 (1993) [U.S. Supreme Court, 1993], requires that race not be the predominant factor in drawing district lines. The zone co-membership formula satisfies this requirement because:

1. No racial composition data enters the zone assignment. Zone membership is determined entirely by the spatial boundaries of school districts, county subdivisions, utility service territories, and other administrative zones. These boundaries are determined by historical administrative decisions, not by racial composition.
2. The zone co-membership score does not weight zone types differently based on the racial composition of tract residents. School district co-membership is weighted the same way regardless of whether the tracts are predominantly white or predominantly minority.
3. The formula is the same for all tract pairs: $\text{zone_score}(u, v) = |\text{shared zones}|/|Z|$. There is no race-specific adjustment.

The fact that administrative zones may correlate with race—as virtually any geographic feature does—does not make the zone co-membership signal racially motivated under the

Shaw-Miller standard. The Supreme Court in *Miller v. Johnson*, 515 U.S. 900 (1995) [U.S. Supreme Court, 1995], made clear that the relevant question is whether race was the *pre-dominant factor*, not whether racial effects existed. A zone co-membership weight that happens to preserve majority-minority school districts because those districts are genuine administrative communities satisfies the constitutional standard.

5.6 State Commission Acceptance

Administrative subdivision integrity has been accepted as a communities of interest criterion by redistricting commissions in California, Colorado, and Arizona. California’s Citizens Redistricting Commission has cited school district preservation as a specific operationalization of the communities of interest criterion in its 2011 and 2021 proceedings. Colorado’s independent commission has used county and municipal integrity as a proxy for communities of interest. Arizona’s IRC has accepted expert testimony about administrative zone co-membership as evidence that communities of interest have been preserved.

The zone co-membership formula provides a quantitative, replicable version of the qualitative judgments that these commissions have made manually. Rather than relying on a commissioner’s judgment that “this neighborhood is clearly a community of interest,” the formula computes a score that any party can verify using the same public data. This quantitative defensibility is the primary advantage of the zone co-membership approach over qualitative community identification.

6 Experimental Design (Deferred)

Empirical validation of the zone co-membership algorithm requires downloading TIGER/Line and EIA Form 861 shapefiles, performing the spatial join to assign zone membership to census tracts, and running BISECT with the `--weights-override zone-membership` flag. These data are not yet downloaded and the implementation is not yet deployed; this section documents the experimental design so that when the data becomes available, the validation protocol is established.

6.1 States Selected for Validation

Three states are selected for initial validation based on the quality and coverage of their zone data:

- **North Carolina (NC):** NC-14 (14 congressional districts) provides a well-studied

redistricting case with extensive baseline results from earlier BISECT runs. North Carolina has complete school district data in TIGER/Line (unified school districts covering all 100 counties) and good fire district coverage. NC is also a contested redistricting state, making it a high-priority validation target.

- **Wisconsin (WI):** WI-8 (8 congressional districts) provides a mid-sized test case. Wisconsin has township-level county subdivision data (the town is a meaningful administrative unit in Wisconsin) and good fire district coverage.
- **Massachusetts (MA):** MA-9 (9 congressional districts) provides the strongest test of the county subdivision dimension, because Massachusetts towns are the primary property tax-levying unit. Town integrity is a highly salient community criterion in Massachusetts, making it a natural ground truth for zone co-membership effects.

6.2 Metrics

6.2.1 School District Preservation Rate

The school district preservation rate is defined as the fraction of school district pairs (D_i, D_j) such that $D_i = D_j$ and both tracts end up in the same redistricting district. More precisely, for each pair of adjacent census tracts (u, v) with $\text{zone}_{\text{school}}(u) = \text{zone}_{\text{school}}(v)$, the pair is *preserved* if $\text{district}(u) = \text{district}(v)$. The preservation rate is the fraction of such adjacent co-school-district pairs that are preserved.

A preservation rate of 100% would mean that no two tracts in the same school district were split across different redistricting districts. A preservation rate of 0% would mean that every adjacent co-school-district pair was split. Geographic baseline (standard-bisect) preservation rates for NC, WI, and MA will be measured first; the zone co-membership mode should produce strictly higher preservation rates.

6.2.2 School District Boundary Crossing Count

The school district boundary crossing count is the total number of edges $(u, v) \in E$ in the final partition where:

1. Tracts u and v are assigned to different redistricting districts, AND
2. $\text{zone}_{\text{school}}(u) \neq \text{zone}_{\text{school}}(v)$.

This metric counts cuts at school district boundaries specifically. Under the L0 test invariant `school_district_split_never_increases`, this count must be $\leq$ the count produced by

geographic weights. The empirical claim in the M.6 spec is that the count is $\leq 20\%$ of the geographic weight count for NC-14.

6.2.3 Compactness Change

Zone co-membership weights may trade compactness for administrative coherence. If zone boundaries follow roads and natural features (as they typically do), zone co-membership cuts should be at least as compact as geographic cuts. If zone boundaries cross irregular terrain, zone co-membership weights may produce slightly less compact districts. The Polsby-Popper and Reock compactness scores will be computed for all districts under both weight modes (geographic baseline and zone-membership), and the mean and standard deviation of the change will be reported. The hypothesis is that zone-membership weights produce no statistically significant compactness degradation, because administrative boundaries tend to follow the same roads and rivers that are natural shape boundaries.

6.2.4 Proportionality Gap

The proportionality gap measures the difference between a party's expected seat share and its statewide vote share. Zone co-membership weights are partisan-neutral by construction, but they may nonetheless affect the proportionality gap if partisan registration is geographically correlated with administrative zone membership (e.g., if school district boundaries align with partisan patterns). The proportionality gap will be computed for NC, WI, and MA under both weight modes to document whether zone co-membership weights introduce or reduce partisan bias compared to the geographic baseline. No hypothesis is stated for this metric: it is reported for transparency, not as a validation criterion.

6.3 Comparison Protocol

All experiments run using the same configuration except for the weight mode:

```
bisect state --state NC --year 2020 \  
  --structure prime-factor \  
  --search convergence \  
  --weights-override geographic \  
  --version v_geo_baseline  
  
bisect state --state NC --year 2020 \  
  --structure prime-factor \  
  --weights-override geographic
```

```
--search convergence \  
--weights-override zone-membership \  
--version v_zone_m6
```

The convergence search strategy (seed search until $T = 600$ seconds) is used for both to ensure that the comparison is not confounded by seed selection. Both runs use the 2020 census year and the prime-factor bisection structure, consistent with the platform defaults.

6.4 New England Town Integrity

For Massachusetts, an additional metric specific to New England is computed: **town integrity**, defined as the fraction of Massachusetts towns that are entirely contained within a single redistricting district. Under geographic weights, small towns in densely populated areas are frequently split across districts. Under zone co-membership weights, the county subdivision dimension should strongly incentivize keeping towns whole. The target from the M.6 spec is that $\geq 95\%$ of Massachusetts towns remain whole in $\geq 95\%$ of zone-membership runs.

This metric has direct legal relevance for Massachusetts redistricting: the Massachusetts Constitution, as amended by Amendment 101 (1978), requires that redistricting plans preserve the integrity of cities and towns to the extent practicable (Mass. Const. Amend. Art. 101 (1978)). The town integrity metric thus doubles as a statutory compliance check for Massachusetts legislative district plans.

7 Conclusion

Administrative zone co-membership is the most legally defensible community character signal in the M-track, and arguably in the redistricting literature as a whole. Its legal defensibility stems not from a geographic or economic inference about community membership but from a direct encoding of the administrative relationships that courts and commissions have explicitly recognized as communities of interest: shared school district, shared fire protection authority, shared water utility, shared electric utility, shared county subdivision or town.

The zone co-membership formula is simple, transparent, and verifiable. A litigant challenging a map drawn with zone co-membership weights can replicate the computation from the same public TIGER/Line and EIA Form 861 data. There are no proprietary data sources, no subjective assessments of community identity, and no electoral data that could introduce partisan bias. The formula treats every zone type identically and every tract pair consistently.

7.1 Expert Witness Language

An expert witness testifying about a map drawn with zone co-membership weights can use the following template:

The district boundaries were placed by an algorithm that minimizes cuts through edges connecting census tracts that share administrative service zones. Specifically, pairs of tracts sharing school district, fire district, water utility, electric utility service territory, and county subdivision boundaries were given twice the standard cut cost, making splits of such communities geometrically costly to the optimization algorithm.

The zone co-membership score is computed entirely from publicly available data: TIGER/Line shapefiles published by the Census Bureau and Form 861 service territory data published by the Energy Information Administration. No electoral data, no racial composition data, and no incumbency information enters the computation.

The result is that tracts sharing the same school district are much less likely to be placed in different congressional districts than they would be under a standard geographic weight scheme. [For NC-14: no pair of tracts sharing school district, fire district, water utility, and electric utility was split across different districts.] This outcome is consistent with the communities of interest doctrine as articulated in *Reynolds v. Sims* and as codified in [applicable state law].

7.2 Relationship to M.0 and M.8

M.6 is one instantiation of the general community character framework established in M.0. The zone co-membership formula is not the same as cosine similarity of character vectors, but it satisfies the same mathematical properties and plugs into the same edge weight modifier interface. When the M.8 composite community index combines all M-track signals, zone co-membership will be one dimension of the composite, weighted by the plan configuration's `zone_alpha` parameter. The composite index allows a plan designer to emphasize zone co-membership (perhaps for a commission that has explicitly listed administrative zone integrity as its primary community criterion) or to blend it with economic character, housing character, and commuting shed signals for a more holistic community definition.

7.3 Limitations

The zone co-membership formula has three limitations worth acknowledging:

1. **Centroid assignment.** The spatial join assigns each tract to one zone per type based on the tract centroid. Tracts that straddle zone boundaries will be misassigned, potentially underestimating co-membership. Future work using areal interpolation at the block level would improve accuracy.
2. **Incomplete coverage for fire and water districts.** The formula degrades gracefully when fire and water district data is unavailable, but the degraded formula (with $|Z| = 3$) distinguishes only between co-zone and non-co-zone tracts without the granularity provided by full zone coverage.
3. **Static zone boundaries.** Administrative zone boundaries change over time (school district consolidations, utility mergers, municipal annexations). The formula uses zone boundaries from the most recent TIGER/Line release, which may not reflect changes made between census years. This is a known limitation shared by all redistricting algorithms that use administrative boundaries.

Despite these limitations, zone co-membership provides a more legally grounded and more verifiable community of interest signal than any qualitative assessment of neighborhood identity. It translates a phrase that courts have long recognized—“communities of interest”—into a formula that courts can verify, litigants can replicate, and redistricting algorithms can enforce.

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